

12(a)	$E = (3.0 \times 10^8)^2 \times 1.66 \times 10^{-27} (= 1.49 \times 10^{-10} \text{ J})$	M1
	$= (1.49 \times 10^{-10}) / (1.60 \times 10^{-19}) = 9.34 \times 10^8 = 934 \text{ MeV}$	A1
	or	
	binding energy = 8.443×95 [or equivalent using La-139 nucleus]	(M1)
	binding energy / mass defect = $(8.443 \times 95) / 0.859 = 934 \text{ MeV}$	(A1)
12(b)(i)	binding energy = $1.865 \times 934 (= 1741.91 \text{ MeV})$	C1
	binding energy per nucleon = $1741.91 / 235$ $= 7.41 \text{ (MeV)}$	A1
12(b)(ii)	less (than)	B1
12(c)	energy = $\{(1.219 + 0.859) - 1.865\} \times 934$ or energy = $(95 \times 8.443) + (139 \times 8.189) - (235 \times 7.412)$	C1
	$= 199 \text{ MeV}$	A1
12(d)	number of reactions = $1.2 \times 10^{-7} \times 6.02 \times 10^{23}$ $= 7.22 \times 10^{16}$	C1
	energy release (for one reaction) = $199 \times 1.60 \times 10^{-13} (= 3.18 \times 10^{-11} \text{ J})$	C1
	power = $(7.22 \times 10^{16} \times 3.18 \times 10^{-11}) / (25 \times 10^{-3})$ $= 9.2 \times 10^7 \text{ W}$	A1